

Question	Marking Guidance	Mark	Comments
02.1	1. Increases/more oxygen dissociation/unloading OR Decreases haemoglobin's affinity for O ₂ ; 2. (By) decreasing (blood) pH/increasing acidity;	2	1. Accept more readily 1. Accept releases more O ₂ 2. Reject if reference made to active site
02.2	1. High(er) affinity for O ₂ (than haemoglobin) OR Dissociates oxygen less readily OR Associates more readily; 2. Allows (aerobic) respiration when diving/at low(er) pO ₂ OR Provides oxygen when haemoglobin unloaded OR Delays anaerobic respiration/lactate production;	2	1. Accept holds O ₂ at lower ppO ₂ 2. Accept acts as an oxygen store
02.3	Correct answer for 2 marks 10.8 to 11 (mins) OR 10 minutes and 48 seconds = 2 marks;; Accept for 1 mark, 10.48 minutes OR Reference to 2057.7 to 2058 (10 700 ÷ 5.2, time oxygen would last if its mass was 1 kg) OR Reference to 56 to 56.3 (10700 ÷ 190, oxygen in 1 kg of seal) OR Reference to 988 (5.2 x 190, oxygen used min ⁻¹ by the seal) OR Incorrect answer with correct answer shown in working	2	
TOTAL		6	

